

1(a)	one mark for each label to correct structure on Fig. 1.1 ; ; ; <i>tonoplast line to anywhere on the vacuolar membrane</i> R if two or more different structures stated for a single label	3
1(b)	ref. to conversion of measured (X-Y) line to μm / AW ; e.g. line is measured in, mm and then multiplied by 1000 / cm and then multiplied by 10 000 evidence that, 60 (± 1 mm) is multiplied by 1000 / 6 cm is multiplied by 10 000 measured diameter is 60 000 μm (and) divide by, magnification / 2000 ;	2
1(c)	<i>two from:</i> rough endoplasmic reticulum ; A rough ER / RER smooth endoplasmic reticulum ; A smooth ER / SER endoplasmic reticulum ; <i>acceptable only if the other structure is <u>not</u> SER / RER</i> 80S / larger, ribosomes ; A 25–30 nm range plasmodesma(ta) ; microtubules ; A microfilaments A cytoskeleton lysosome(s) ; Golgi (body / apparatus / complex) ; secretory / Golgi, vesicles ; AVP ; ; e.g. chromatin EM detail of chloroplast EM detail of mitochondrion A mitochondrion nuclear pore nuclear envelope R nuclear membrane	2

Question	Answer	Marks
1(d)(iii)	<i>two from:</i> transpiration occurs ; <i>in context of via stomata or via cuticle</i> diffuses / moves / AW, to, (external) environment / atmosphere / outside / out of the leaf / from leaf surface ; R evaporation / osmosis R from cell surface via (open) stomata (for, gas exchange / carbon dioxide entry / photosynthesis) ; <i>accept Ψ for water potential</i> I solute potential / ref. to concentration (movement of water vapour) down water potential gradient ; A down water vapour potential gradient A high to low water potential / to lower water potential / from higher water potential R from high to low water potential <u>gradient</u>	2

Question	Answer	Marks
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